

Hw3 con.

4. Proof that min is associative (in the case of real #'s)
by breaking it into cases:

Prove $\min(\min(a, b), c) = \min(a, \min(b, c))$ for all real numbers a, b, c

Case 1: $a \leq b$ and $a \leq c$

$\begin{matrix} b/c \\ \uparrow \\ a \leq c \end{matrix}$

$\min(a, b) = a$. So left side $\min(\min(a, b), c) = \min(a, c) = a$.

The right side: $\min(b, c) \geq a$, so $\min(a, \min(b, c)) = a$

→ Both sides equal a .

Case 2: $b \leq a$ and $b \leq c$

$\begin{matrix} a/c \\ \uparrow \\ b \leq c \end{matrix}$

$\min(a, b) = b$, so left side $\min(\min(a, b), c) = \min(b, c) = b$.

Right side: $\min(a, \min(b, c)) \rightarrow \min(a, b) = b$.

→ Both sides equal b .

Case 3: $c \leq a$ and $c \leq b$

$\min(a, b) \geq c$, so left side $\min(\min(a, b), c) = c$.

Right side: $\min(a, \min(b, c)) = \min(a, c) = c$

→ Both sides equal c .

→ Both sides equal the smallest value.

Three cases cover all possibilities, and in every case the left and right side equal each other.

Therefore, for all real numbers (a, b, c) the min is associative.